



Caribbean Coral Reef Degradation Policy White Paper

Prepared for Coral Reef Alliance

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December 9, 2018

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PPD 554

EXECUTIVE SUMMARY

Coral reef ecosystems in the Caribbean face high rates of degradation. With the loss of coral biomass and reef diversity comes economic instability, food insecurity, and increased coastal vulnerability. Coral reef degradation is a result of unsustainable coastal development and chemical waste runoff, carbon emissions and climate change, and overfishing and harmful fishing practices. Due to the limited time frame for addressing this issue, stakeholders must begin immediately to mitigate reef degradation, promote positive environmental impact, and provide for the regrowth and redevelopment of coral reef ecosystems. The recommended course of action is a comprehensive approach combining training and support of local fishermen as coral gardeners and small business owners, criteria index updates for permitting purposes, implementation of a fisheries co-management model, and a targeted and efficient clean energy campaign.

ISSUE OVERVIEW

Since the 1980s, Caribbean coral reef degradation has occurred at increasingly higher than natural rates and to an increasingly greater than natural extent (Aronson et al., 2002). Individual coral reef mass-mortality events have also increased in frequency and magnitude (Roff, Zhao, & Mumby, 2015). Reef degradation, defined as “sustained and ongoing habitat [loss] in coral reef ecosystems,” applies the loss of coral reef biomass and diversity not only to the coral themselves but also to reef fish and other organisms that live in the reef ecosystem (Pratchett, Hoey, & Wilson, 2014, p. 37). Absolute coral cover in the Caribbean fell from about 50% in the 1970s to about 10% in 2003, with a -5.46% mean annual rate of change over the same period (Gardner et al., 2003). Furthermore, forecasts show increasingly negative carbonate budgets, which measure the difference between the production and erosion of coral carbonate and highlight trends in coral growth or degradation (Kennedy et al., 2013). These projections indicate total functional collapse of Caribbean reef ecosystems by 2080 assuming no action is taken to reduce rates of decay (Kennedy et al., 2013). A 16.5% chance exists that coral reefs are so degraded they will be unable to reproduce successfully (Brander et al, 2012).

Harms and Magnitude

1. Economic Instability

As coral reefs die off, local human populations incur much of the costs through economic decline and instability. Coral reefs provide global goods and services valued at around \$30 billion annually (Chen et al., 2015). This is reflected in *Figure 1* below, where the global average value of coral is measured using population and income growth, GDP per capita, the number of international visitors, and other factors (Brander et al., 2012). These include savings and income generated by quasi-public goods and non-market services such as reef-based tourism and recreation, fish-catch exports and trade, building materials, agriculture, and coastal protection (Chen et al., 2015; Allison et al., 2009). In the Caribbean, economic losses due to coral reef degradation cost between an estimated \$350 and \$870 million per year (Chen et al., 2015; Burke and Maidens, 2004). The negative effect on national economies is most evident in the decline of

“climate-sensitive” industries such as agriculture, fisheries and trade, and tourism (Allison et al., 2009).

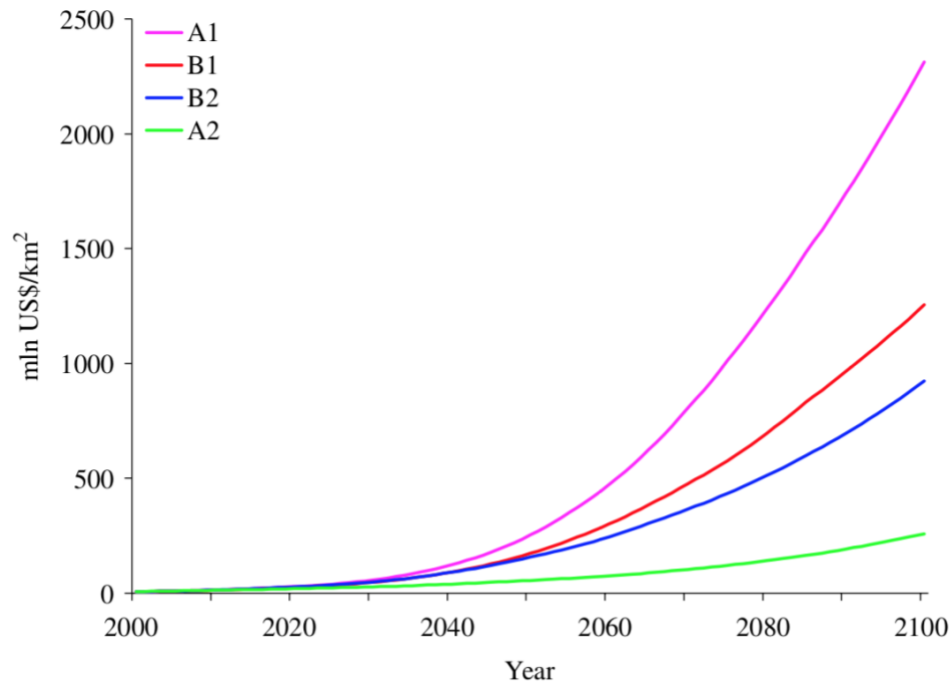


Figure 1. Value of Coral. This graph shows the global average value per km² of coral reef area. Values are calculated for the four marker scenarios in the IPCC Special Report on Emission Scenarios. (Brander et al., 2012; Nakicenovic and Swart, 2001).

2. Food Insecurity

Coral reef degradation further yields high rates of food insecurity. Coral reefs are the most diverse marine habitat per unit area and the most diverse marine habitat overall (Knowlton et al., 2010). Estimates of biodiversity in coral reef ecosystems range from 500,000 to 10 million species (Knowlton et al., 2010). For Caribbean local communities, reef fish contribute between 34 and 76% of protein intake from animals (McClanahan, Allison, & Cinner, 2015). As coral reef ecosystems decline, so too do the abundance and species richness of fish available in reef-based fisheries (Wilson et al., 2006). *Figure 2* shows the change in abundance of 55 species of reef fish in response to a decline in coral cover (Wilson et al., 2006; *Appendix A*). More than 50% of species observed significantly declined in abundance when net coral cover declined (Wilson et al., 2006). *Figure 3* below shows the relationship between coral cover decline and species richness of reef ecosystems (Wilson et al., 2006). With a strong absolute correlation coefficient ($r = 0.61$), species richness and net coral cover decline exhibited a negative relationship (Wilson et al., 2006). The declines in abundance and species richness, in addition to the economic instability and loss of income that result from coral reef degradation, reduce access and meal diversity for local communities, leading to high rates of food insecurity.

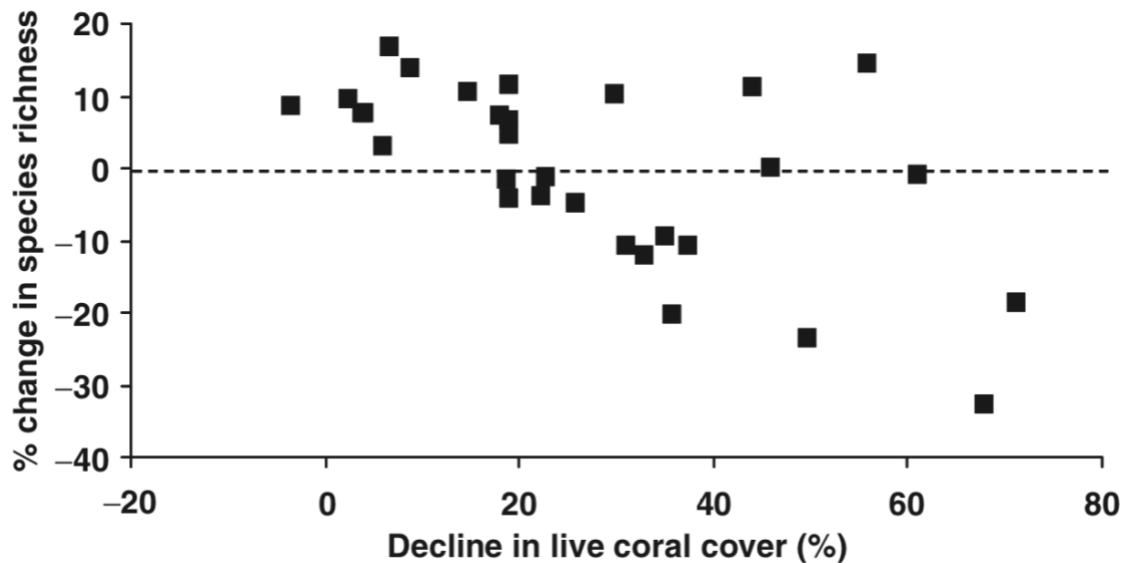


Figure 3. Relationship between species richness and coral cover decline. This graph shows a negative relationship between declining coral cover and species richness of coral reef ecosystems (Wilson et al., 2006).

3. Coastal Vulnerability

Coastal communities are more vulnerable to natural disasters as a result of coral reef degradation. Coral reefs provide natural infrastructure that significantly reduces the risk of coastal damage from waves and storms (Ferrario et al., 2014). Coral reefs reduce wave energy, including that of storm waves, by an average of 97% (Ferrario et al., 2014). Reefs reduce wave height by an average of 82% (Ferrario et al., 2014). As shown in *Figure 4* and *Figure 5*, the reef crest accounts for 83% of wave energy attenuation and 61% of wave height reduction (Ferrario et al., 2014; *Appendix B*).

The coastal protection services provided by coral reefs yield a coastal protection value of an estimated \$9 billion globally (Cesar, Burke, & Pet-Soede, 2003). The coastal protection value (CPV) is an indirect-use economic valuation of the non-market coastal protection services provided by coral reefs that is derived from property values, land use values, employment and wages, and other economic indicators at risk of loss from natural disaster (van Zanten, van Beukering, & Wagtendonk, 2014). The CPV can also be understood as the economic value of reef-based coastal protection at risk given coral reef degradation. In the Caribbean, the CPV is estimated between \$700 million and \$2.2 billion per year (Burke and Maidens, 2004).

Causes of Coral Reef Degradation

1. Carbon Emissions and Climate Change

Oceans absorb an estimated 30% of anthropogenic carbon dioxide emissions (Ricevuto, Vizzini, & Gambi, 2015). Once absorbed, carbon dioxide dissolves into an acid that alters seawater chemistry and lowers pH, a process known as “ocean acidification” (Brander et al., 2012; Caldeira and Wickett, 2005). As seawater chemistry in shallow environments is highly variable, coral reef ecosystems are highly susceptible to ocean acidification (Cinner et al., 2016). Ocean acidification yields high rates of carbonate dissolution, or the destruction of coral carbonate, and is expected to reduce marine calcification, or the production of coral carbonate, by up to 50% by 2050 (Andersson, Bates, & Mackenzie, 2007; Cinner et al., 2016). The aggregate effect of ocean acidification on carbonate budgets, considering both carbonate dissolution and marine calcification, results in net erosion of coral cover (Hoegh-Guldberg et al., 2017). Ocean acidification, and the consequent coral reef degradation, is directly related to human pollution, with current atmospheric carbon dioxide composition reflecting a more than 39% growth from pre-industrial levels (Leung, Caramanna, & Maroto-Valer, 2014). The anthropogenic emissions leading to this atmospheric change come from combustion of fossil fuels and biofuels, mostly used in power generation and industry, transportation, farm operations, and commercial development (Bond et al., 2004; Lal, 2004).

2. Coastal Development and Waste Runoff

A strong negative correlation exists between levels of coastal commercial development and levels of coral reef degradation in the Caribbean (Bozec et al., 2008). Coastal development is also related to fish functional diversity, algal coverage, and substrate coverage in reef ecosystems (Bozec et al., 2008). This is due to the terrestrial runoff into coastal waters associated with development (Fabricius, 2005). Waste runoff from development increases the concentration of dissolved inorganic nitrogen (DIN) and other nutrients in coral-rich waters (Fabricius, 2005). DIN acts as a fertilizer for zooxanthellae, increasing densities of the microorganism that normally lives in a balanced symbiotic relationship with coral (Fabricius, 2005). As zooxanthellae densities increase, carbon dioxide intake by zooxanthellae increases and carbon dioxide availability in the water decreases (Fabricius, 2005). With less carbon dioxide present, coral calcification decreases (Fabricius, 2005). Per the definition of a carbonate budget, the decrease in coral calcification resulting from waste runoff leads to increased rates of coral reef degradation (Kennedy et al., 2013). The relationships between coastal development and reef outcomes are expressed in *Figure 6* (Bozec et al., 2008; *Appendix C*).

Coastal development also furthers coral reef degradation by contributing to ocean acidification and coral bleaching through development-related carbon emissions and particulate sedimentation of reef waters (Bond et al., 2004; Campbell, 2017). Carbon emissions, released throughout the development process, cause oceans to acidify, reducing carbonate budgets and increasing reef degradation (Hoegh-Guldberg et al., 2017). Particulate sedimentation in reef waters, caused by waste runoff, reduces light penetration to zooxanthellae, impeding photosynthesis and leading to coral bleaching (Campbell, 2017). Coral bleaching, or the expulsion of zooxanthellae by coral, is an indicator of reef degradation (Fabricius, 2005).

3. Overfishing and Harmful Fishing Practices

Caribbean coral compete with sponges and macroalgae for biomass dominance in the reef ecosystem (Mumby, 2016). In a balanced ecosystem, parrotfish and other herbivorous fish protect coral from algal overgrowth (Mumby, 2016). Where parrotfish habitats are unregulated by conventional instruments, such as no-take zones, overfishing by commercial and recreational fishers has depleted parrotfish populations to unsustainable rates (Mumby, 2016). Overfishing has implications on coral reef degradation (Loh et al., 2015). Overfished sites in the Caribbean were found to have three times as much algal overgrowth and sponge-to-coral contact rates of 25.6%, as compared with 12.0% at less-fished sites (Loh et al., 2015). This is significant, as algal growth is strongly negatively correlated with coral growth (Mumby & Harborne, 2010). Algal overgrowth in reef ecosystems limits light exposure to coral and reduces photosynthetic potential of coral symbionts, leading to excess coral bleaching (Pawlik et al., 2007). The relationship between algal overgrowth and reef degradation, measured by structural complexity of reef ecosystems, is expressed in *Figure 7* below (Graham and Nash, 2012).

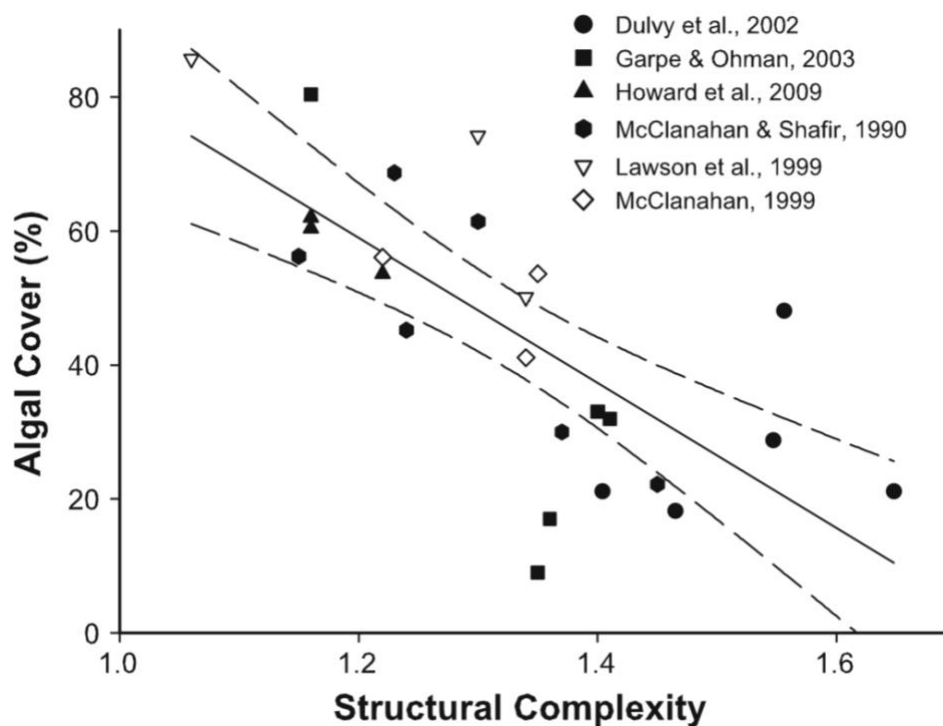


Figure 7. Relationship between algal cover and structural complexity. This graph shows a negative relationship between algal cover and structural complexity of coral reef ecosystems, a measure of reef degradation (Graham and Nash, 2012).

Beyond overfishing, harmful fishing practices contribute to coral reef degradation (Anthony et al., 2015). Harmful fishing practices include bomb fishing and cyanide fishing (Anthony et al., 2015). These methods cause damage to the physical environment of coral reefs, along with the loss of biodiversity through poisoning and destruction (Calado et al., 2014). Cyanide fishing, for

example, stuns the targeted fish, making them easier to collect, but kills coral, invertebrates, and other organisms immediately (Calado et al., 2014). Cyanide fishing is often specifically used by fisherman for coral reef sites, as it stuns fish in hard-to-reach places, causing them to float to the surface (Calado et al., 2014).

ANALYSIS

Coral Reef Alliance (CORAL) is a non-profit organization based in Oakland, California that operates in coastal areas around the world. As CORAL is a non-profit organization, it secures most of its funding through government grants and donations. CORAL has no legislative power but has strong relationships in at-risk regions, including the Caribbean. CORAL has in the past run effective conservation programs including a USAID sponsored effort to foster sustainable tourism in the Mesoamerican Barrier Reef System, formulation of a World Heritage Site in Kiribati, and creation of the first anchor-free zone in Fiji. CORAL is well-situated to address coral reef degradation in the Caribbean.

Goals and Objectives

Effectively combatting Caribbean coral reef degradation will take an efficient and coordinated effort of policymakers, academics, civil society, and organizations like CORAL at the local, regional, and global levels. The overarching goal must be to mitigate coral reef degradation, promote positive environmental impact, and provide for the regrowth of coral reef ecosystems. Objectives to achieve this goal are:

1. Educate and mobilize local citizenry
2. Reduce carbon emissions
3. Encourage sustainable coastal development and management

Political Context and Policy Window

Many actors are working to fight coral reef degradation. The timeframe for such action, however, is limited. Carbonate budget projections considering local, regional, and global stressors indicate full functional collapse of Caribbean reef ecosystems by 2080 (Kennedy et al., 2013). A 16.5% chance exists that coral reefs are so degraded they will be unable to reproduce successfully (Brander et al., 2012). A policy window is open, but efforts must be coordinated immediately to ensure maximum possible impact. In this endeavor, CORAL can lead the way.

Political support and access to conservation funds abound. In the United States, the 115th Congress passed the Department of the Interior, Environment, and Related Agencies Appropriations Act for Fiscal Year 2019 in the House of Representatives and Senate, authorizing \$91 million for coral reef conservation activities, among other things (H.R. 6147, 2018). The bill passed with different amendments in the two bodies, which are now working in conference to resolve their differences (Library of Congress, 2018). Another bill, the Coral Reef Conservation Reauthorization Act of 2018, would provide an additional \$16 million for coral reef conservation activities (H.R. 5996, 2018). The Coral Reef Conservation Act of 2000, which H.R. 5996 would reauthorize should it become law, established the Coral Reef Conservation Program to direct United States coral conservation efforts (Coral Reef Conservation Act, 2000).

Civil society in the Caribbean is also trending toward conservation. In 2017, a Belizean moratorium on oil exploration and drilling in the Belize Barrier Reef Reserve System, the coral reef system spanning the nation's coast, became law (Root, 2018). The moratorium was expanded as law following a years-long effort to place a citizens' referendum on the ballot in Belize (Root, 2018). A similar ban exists along the coast of the Yucatan, Mexico's Caribbean peninsula (Root, 2018).

Stakeholders

Stakeholders include those who are affected by coral reef degradation and those who have a vested interest in combatting and reversing reef degradation. Globally, there are 197 million people who live below 10 meters elevation and within 50 kilometers of a coral reef ecosystem and 100 million people who live below 10 meters elevation and within 10 kilometers of a coral reef ecosystem (Ferrario et al., 2014). There may be as many as 22 million people who meet these conditions in the Caribbean (Ferrario et al., 2014). These people receive economic, health, and safety benefits from reefs. So, too, do the nearly 20 million tourists who contribute on average 15% of GDP in each Caribbean country (Lorde, Li, & Airey, 2016). Tourists contribute as much as 84.1% of GDP in Antigua and Barbuda (Lorde, Li, & Airey, 2016).

Other stakeholders include fishers and trade partners, who benefit from healthy reef fisheries (Burke et al., 2011). The ornamental species trade, which includes reef fish kept as pets, is valued between \$200 million and \$330 million globally (Burke et al., 2011). The live reef food fish trade had an estimated value of \$810 million in 2002 (Burke et al., 2011). Finally, stakeholders include policymakers, industry, and environmentalist non-profit organizations. These are the people and organizations tasked with mitigating, preventing, and reversing coral reef degradation.

Evaluative Criteria

In selecting alternatives, several evaluative criteria were chosen. These include: effectiveness, efficiency, equity, and feasibility. Effectiveness assesses the proposal's furtherance of the goal of mitigating reef degradation, promoting positive environmental impact, and providing for the regrowth of reef ecosystems. Efficiency measures the proposal's benefits against its costs. Equity assesses the proposal's distributive justice. Feasibility considers political conditions and implementation complexity. Combined, these evaluative criteria ensure the proposed solution accounts for all aspects of the effort to combat reef degradation, including stakeholders, goals, and potential obstacles. The solutions are evaluated against the criteria using a criteria-alternative matrix (*Appendix E*).

Proposed Alternatives

1. Coral Gardener Training and Support Program

CORAL should establish a coral gardener training and support program. Coral gardening is a reef restoration method whereby a limited amount of coral tissue and skeleton are extracted from

healthy reefs and cultivated within in situ or ex situ nurseries, before being transplanted to degraded reefs (Rinkevich, 1995; Young, Schopmeyer, & Lirman, 2012). A coral gardener training and support program would train local fishermen to serve as coral gardeners, empowering them to supplement their income by establishing ecotourism companies that give tourists the opportunity to partake in coral restoration efforts. This would address reef restoration as well as reduce overfishing. Fishermen would be provided with SCUBA training, coral gardening education, and business services including marketing and pricing.

Effectiveness

Coral gardening ranked as the most effective reef restoration method in a survey administered to professional practitioners (Young, Schopmeyer, & Lirman, 2012). Coral gardening facilitates the regrowth and reintroduction of healthy coral into degraded reefs (Rinkevich, 1995). On overfishing and unsustainable fishing practices, estimates regarding the mobilization of fishermen show that each fisherman trained as a coral gardener keeps 12.5 pounds of parrotfish per day on the local reefs (Lirman and Schopmeyer, 2016). Additionally, ecotourism businesses run by fisherman-gardeners would augment earned income and improve local economies.

Efficiency

Coral gardening is low-cost and technically simple (Young, Schopmeyer, & Lirman, 2012). This makes it easy for fisherman-gardeners to conduct coral gardening operations, ensuring increased profits as they keep costs low. CORAL itself incurs only the cost of training fishermen and supporting them with pro-bono consulting services. Empowering local citizenry would mobilize an untapped work force and reduce labor and overhead costs on CORAL. If fisherman-gardeners are further trained to provide training to other locals, coral restoration efforts may spread without influence by CORAL.

Equity

A coral gardener training program maximizes equity by engaging local citizenry and offering economic improvement opportunities. Equity is further created through restoration of coral reef ecosystems and support of healthy fisheries, both quasi-public goods that provide non-market services attributable to all stakeholders.

Feasibility

The coral gardener training program requires low commitment from local, regional, and federal governments. With the establishment of ecotourism businesses operated by fisherman-gardeners, governments will be responsible for providing business permits. CORAL can leverage relationships to secure favorable permitting requirements.

The coral gardening training program is low-cost, requiring CORAL to provide only training and support of local fisherman-gardeners. The newly trained fisherman-gardeners will be expected to finance the operation of their own ecotourism businesses as well as their own coral gardening

efforts. Local buy-in from fishermen to participate in this program may be the only hurdle that needs to be overcome.

2. Coastal Development Permitting Criteria Updates

CORAL should work with local and regional government partners to update criteria indices for coastal development and land use permitting. These may include updates to carbon emissions standards, waste management standards, and chemical use allowances in accordance with the United Nations Sustainable Development Goals. A comprehensive research study on the most impactful stressors of coastal development and land use should be administered with a specific focus on carbon emissions, chemical runoff, sedimentation, proximity to at-risk areas, and overall environmental disturbance.

Effectiveness

Enacted permitting requirements will regulate development and reduce stressors on coral reef ecosystems. Though permitting requirements already exist, updates will reflect changing issues. As developers must adhere to government-imposed regulations, permitting requirements would face high rates of compliance. This solution would address pollution, sedimentation, and the impact of coastal development.

Efficiency

While this solution is relatively low-cost, it will require investment in research and lobbying activities to ensure proper implementation. Additionally, extra investment may be necessary as permitting processes are not streamlined across the Caribbean (Sutton-Grier, Wowk, & Bamford, 2015). CORAL would have to develop individual proposals for each local government.

Equity

Equity concerns exist as the implementation of stricter permitting requirements disenfranchises developers. CORAL can expect developers to outspend the organization in reverse lobbying efforts to ensure they are not faced with increased regulation.

Feasibility

The imposition of expanded regulations on development, which brings capital investment to local communities, will likely lack political support. In addition to lobbying competition from developers, CORAL can expect to face policymakers who are more inclined to satisfy the developers. Implementation, however, is technically feasible. CORAL would conduct or sponsor a research study and work with officials to incorporate results into law. With the availability of U.S. grants, research costs can likely be heavily subsidized (H.R. 6147, 2018).

3. Fisheries Co-management

CORAL should establish a fisheries co-management system that engages local resource users, local governments, and CORAL managers. This system would create fisheries management boards incorporating all three stakeholders, ensuring community buy-in and government support. Local resource users, through these fisheries management boards, would clearly define common-pool boundaries, establish collective choice rules, and create mutually agreed upon graduated sanctions scales. Co-management, rather than CORAL or government managed no-take marine protected areas, would legitimate management in the eyes of the community and ensure community compliance (Cinner et al., 2012).

Effectiveness

Co-managed fisheries ensure increased biomass survival and fish abundance (Cinner et al., 2012). Increased biomass combats algal overgrowth, providing for the rebalancing of reef ecosystems (Cinner et al., 2012). *Figure 8* shows biomass outcomes in fisheries with no-take policies, co-managed systems, and no local management (*Appendix D*). While outcomes in no-take fisheries appear to be higher than those in co-managed fisheries, the difference is not statistically significant (Cinner et al., 2012; *Appendix D*). Further, co-managed fisheries yield high compliance rates due to local buy-in (Cinner et al., 2012).

Efficiency

Costs of managing fisheries are minimized by co-management, given the shared responsibility of CORAL, local resource users, and local government. Additionally, revenue is earned from sanctions on those that misuse fisheries, as agreed upon through collective choice.

Equity

Co-managed fisheries increase equity by providing resource users with a greater say in the allocation of resources (Cinner et al., 2012). Additionally, the use of a co-management model over a no-take model allows fisheries related economic activity to continue unabated. Equity concerns do exist though, as benefits tend to favor wealthier resource users (Cinner et al., 2012). These people tend to live closer to fisheries and are more capable of paying sanctions (Cinner et al., 2012).

Feasibility

Co-management models earn wide support from local communities, who express a perceived livelihood benefit from co-managed fisheries (Cinner et al., 2012). Co-management allows communities to continue to make use of natural resources, while at the same time feeling as if they are contributing to the betterment of the environment. Implementation is made feasible through the shared-cost nature of co-managed fisheries.

4. Clean Energy Campaign

CORAL should invest in clean energy research and education. A clean energy campaign would mobilize local communities to transition to clean energy. This may include working with

community leaders to organize clean energy ballot measure campaigns, gather signatures, and lobby for local regulation. CORAL should create stratified education programs to address energy users in different sectors of society, including corporate users, schools, households, and government.

Effectiveness

Clean energy campaigns have proven to be very effective. Data show that household-level behavioral changes can reduce carbon emissions by up to 44% after 10 years (Dietz et al., 2009). With stratified education programs, CORAL's campaign can reach many stakeholders and ensure greater energy outcomes.

Efficiency

A clean energy campaign, especially one that addresses so many parts of society and is administered across the region, faces high financial costs and low response yield. CORAL would have to invest much in time, money, and political negotiation to ensure an effective campaign. Responses by society may be low and effects may not be felt for a long time.

Equity

A stratified clean energy campaign increases equity by addressing all energy users. By targeting different sectors of society, CORAL would spread the effects of its campaign. Equity is reduced, however, as costs of transitioning to clean energy are high on poor energy users.

Feasibility

As a stratified clean energy campaign seeks to address multiple sectors, access and scope may be limited in implementation. Companies or governments may refuse to admit CORAL educators, and CORAL will likely be unable to administer its campaign region-wide. Additionally, associated costs are high and payoff is low.

RECOMMENDATION

A comprehensive approach to combatting coral reef degradation is recommended. Such an approach would combine the alternative solutions of a coral gardener training program, coastal development permitting updates, fisheries co-management, and a clean energy campaign. By combining these alternatives, CORAL will effectively approach the goal of mitigating coral reef degradation, promoting positive environmental impact, and providing for the regrowth of coral reef ecosystems. This goal will be attained by meeting the objectives of empowering local citizenry, reducing carbon emissions, and encouraging sustainable coastal development and management.

Implementation Considerations

In order to efficiently implement a comprehensive approach, CORAL should strengthen relationships with governments and invest early in research and education. Action should be taken only after a meticulous planning period, as local and regional approaches yield different cost constraints. Based on the criteria-alternative matrix, CORAL should prioritize the coral gardening approach first, followed by fisheries co-management, then permitting criteria updates, and finally a long-term clean energy campaign (*Appendix E*).

CONCLUSION AND NEXT STEPS

Coral reef degradation is advanced and nearing the point of no return (Kennedy et al., 2013; Brander et al., 2012). Overall degradation and mass-mortality events have increased in frequency and magnitude since the 1980s (Aronson et al., 2002; Roff, Zhao, & Mumby, 2015; Gardner et al., 2003). CORAL should enact a comprehensive approach to mitigate coral reef degradation, promote positive environmental impact, and provide for the regrowth of coral reef ecosystems. Next steps include development of relationships with local and regional actors, financial and logistical planning, and creation of a coral gardener training program. CORAL can secure funds from grants associated with the Coral Reef Conservation Program of the National Oceanic and Atmospheric Administration.

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APPENDIX A: FOOD INSECURITY

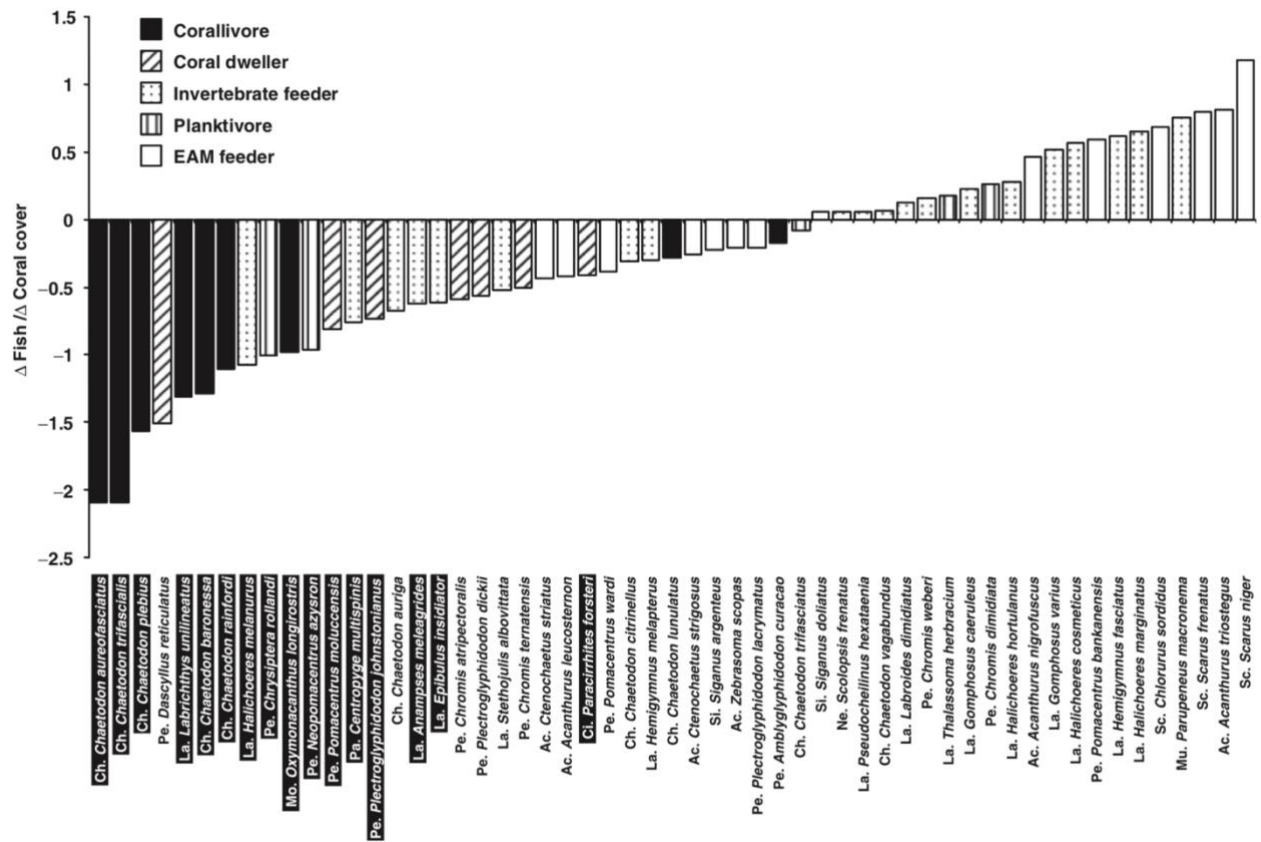


Figure 2. Change in abundance of reef fish. This graph shows the change in abundance of 55 species of reef fish in response to a decline in coral cover. Names highlighted indicate significant consistency of response across four locations observed. (Wilson et al., 2006).

APPENDIX B: COASTAL VULNERABILITY

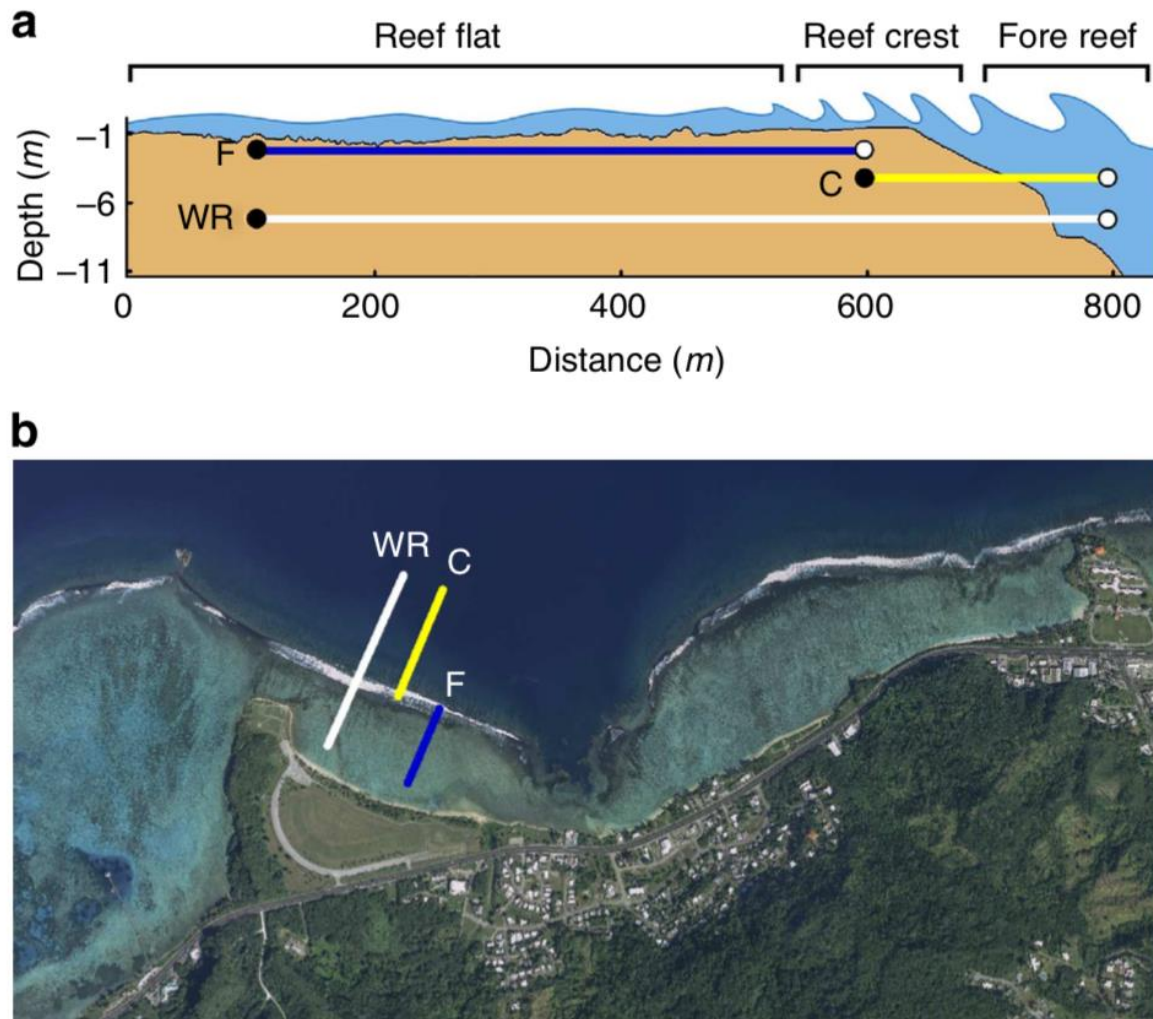


Figure 4. Reef structure for wave attenuation. This image shows the structure of a reef with application to its ability to reduce the energy and height of coastal waves. (Ferrario et al., 2014).

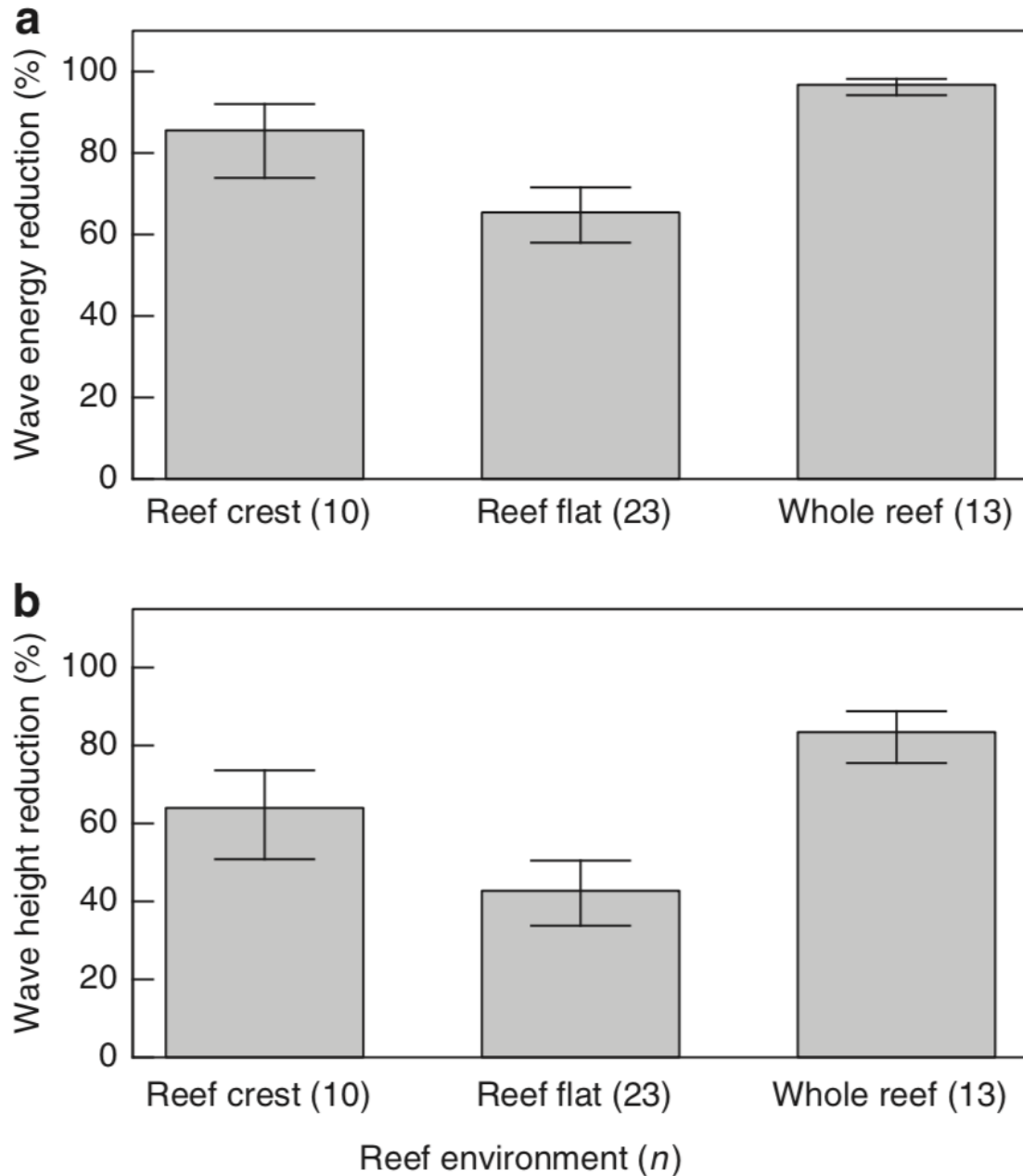


Figure 5. Wave attenuation by reef component. This graph attributes wave attenuation levels to each reef component, as pictured in Figure 4. The reef crest, as opposed to the reef flat, is most associated with attenuation. The whole reef reduces wave energy by as much as 97% and wave height by as much as 82%. (Ferrario et al., 2014).

APPENDIX C: COASTAL DEVELOPMENT AND WASTE RUNOFF

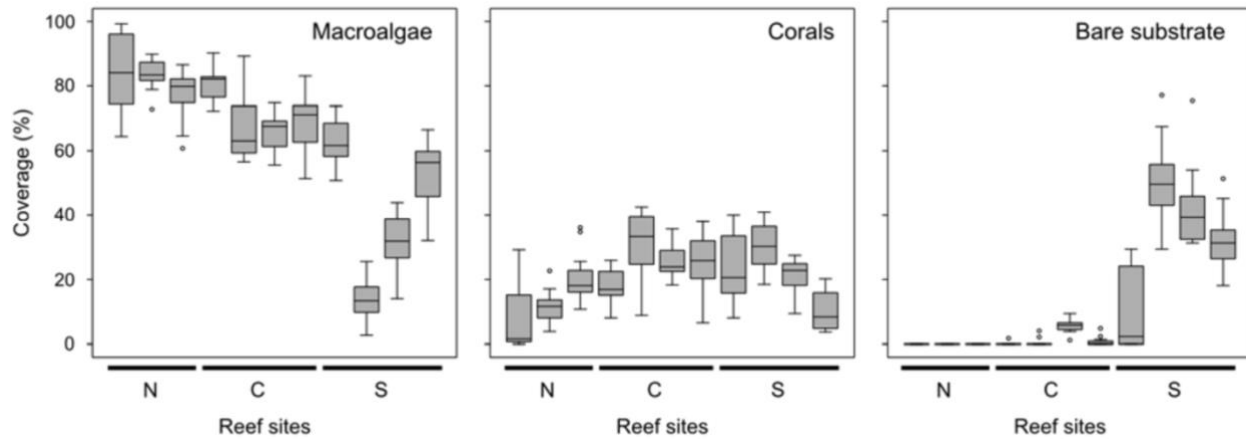


Figure 6. Effects of coastal development. This three-panel graph shows the effects of coastal development on the cover of macroalgae, corals, and bare substrate at three different sites along the Yucatan coast of Mexico. The data yield a significantly greater amount of macroalgae where coastal development is greater and a significantly less amount of coral where coastal development is greater. The sites are coded as follows:

1. N=Riviera Maya, heavy coastal development
2. C=Sian Ka'an Biosphere Reserve, low human density and protected status
3. S=Costa Maya, low human density, small but rising tourism development

(Bozec et al., 2008).

APPENDIX D: FISHERIES CO-MANAGEMENT

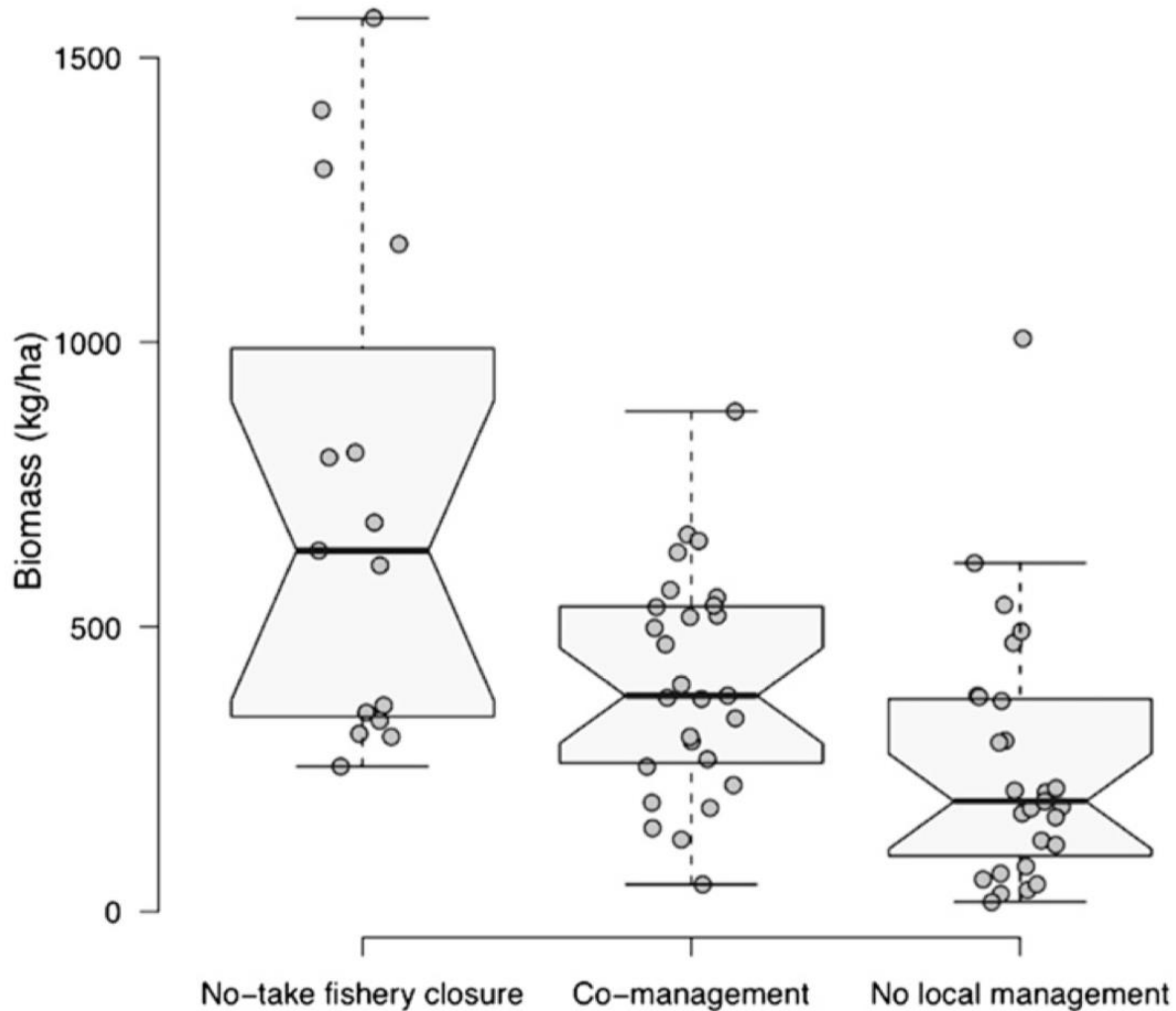


Figure 8. Biomass present in fisheries. This chart shows the abundance of biomass available for fishing at fisheries with no-take policies, co-management structures, and no local management. While no-take fisheries have the most biomass available, indicating coral health, the difference is not statistically significant. Co-management allows for high abundance while also continuing to allow fishermen to take at an appropriate level. (Cinner et al., 2012).

APPENDIX E: CRITERIA-ALTERNATIVE MATRIX

1=Most desirable, 4=Least desirable

	Effectiveness	Efficiency	Equity	Feasibility	Naïve Ranking
Training local fishermen	Addresses reef restoration, overfishing, economic instability	Low-cost, technically simple	Empowers local community, addresses stressors	Politically, technically, and financially feasible	1
Updating permitting criteria indices	Addresses pollution, sedimentation, impact of development	Investment in research and lobbying	Disenfranchises developers	Politically challenging, technically feasible	3
Fisheries comanagement	Addresses overfishing	Low-cost, high compliance	Benefits wealthy	Politically, technically, and financially feasible	2
Clean energy campaign	Addresses pollution	High cost, low effect	Disenfranchises poor	Access and scope challenges, high costs	4